

Chapter 7 and Appendix in “Power System Economics: The Nordic Electricity Market, Ivar Wangensteen”.

$$C_i = C_i(P_{Gi})$$

$$c_i = \frac{\partial C_i}{\partial P_{Gi}} = c_i(P_{Gi})$$

$$B_j = B_j(P_{Lj})$$

$$d_j = \frac{\partial B_j}{\partial P_{Lj}} = d_j(P_{Lj})$$

$$\max f = \sum_{j=1}^m B_j - \sum_{i=1}^n C_i = \max \left[\sum_{j=1}^m \int_0^{P_{Lj}} d_j(P) dP - \sum_{i=1}^n \int_0^{P_{Gi}} c_i(P) dP \right]$$

Constant demand (price inelastic): cost minimization:

$$\max f = \left[Const - \sum_{i=1}^n \int_0^{P_{Gi}} c_i(P) dP \right] \Rightarrow \min \left[\sum_{i=1}^n \int_0^{P_{Gi}} c_i(P) dP \right]$$

Optimization subject to all constraints, among them the load flow equations:

$$g(\mathbf{x}, \mathbf{P}) = 0$$

$$\mathbf{P} = \begin{bmatrix} P_{G1} \\ \cdot \\ \cdot \\ P_{Gn} \\ P_{L1} \\ \cdot \\ \cdot \\ P_{Ln} \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} \delta_1 \\ \cdot \\ \cdot \\ \delta_{n+m} \\ |U_1| \\ \cdot \\ \cdot \\ |U_{n+m}| \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} U_1 \\ \cdot \\ \cdot \\ U_{n+m} \end{bmatrix}$$

In principle voltage angles and magnitudes at all buses.

In practical approaches $\mathbf{x}$ depends on the way the problem is formulated, e.g. AC load flow, DC flow, and the way the buses are defined (slack, PV, PQ). Here treated a little "loosely" to explain the general concept, then more specifically in a DC grid (voltage = voltage magnitude) and afterwards DC load flow in AC grid (angles)).

$$\text{Lagrangian : } L(\mathbf{x}, \mathbf{P}, \boldsymbol{\lambda}) = f(\mathbf{P}) + \boldsymbol{\lambda}^t \mathbf{g}(\mathbf{x}, \mathbf{P})$$

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \cdot \\ \cdot \\ \lambda_{n+m} \end{bmatrix}$$

First order KKT conditions:

$$\frac{\partial L}{\partial \boldsymbol{\lambda}} = 0 \Rightarrow \mathbf{g}(\mathbf{x}, \mathbf{P}) = 0$$

$$\frac{\partial L}{\partial \mathbf{x}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^t \boldsymbol{\lambda} = 0$$

$$\frac{\partial L}{\partial \mathbf{P}} = 0 \Leftrightarrow \frac{\partial f}{\partial \mathbf{P}} - \boldsymbol{\lambda} = 0 \Rightarrow \boldsymbol{\lambda} = \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

where $\mathbf{c}$ is vector of marginal generation costs and $\mathbf{d}$ vector of consumers' marginal willingness to pay.

With fixed demand this is again the well-known result that in the optimum all generators have the same marginal cost unless they are at min or max.

Add constraints on transfer capacity: $\mathbf{h}(\mathbf{x}) \leq 0$

and generation capacity: $P_{Gi} \leq P_{i,max}$

$$L(\mathbf{x}, \mathbf{P}) = f(\mathbf{P}) + \boldsymbol{\lambda}^t \mathbf{g}(\mathbf{x}, \mathbf{P}) + \boldsymbol{\eta}^t \mathbf{h}(\mathbf{x}) + \boldsymbol{\gamma}^t (\mathbf{P} - \mathbf{P}_{\max})$$

First order KKT conditions:

$$\mathbf{g}(\mathbf{x}, \mathbf{P}) = 0 \quad \lambda$$

$$\mathbf{h}(\mathbf{x}) = 0 \quad \eta$$

$$(\mathbf{P} - \mathbf{P}_{\max}) = 0 \quad \gamma$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^t \lambda + \left(\frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right)^t \eta = 0$$

$$\lambda = \begin{bmatrix} \mathbf{c} + \gamma \\ \mathbf{d} \end{bmatrix}$$

with fixed demand:

$$\lambda = \begin{bmatrix} \mathbf{c} + \gamma_1 \\ \gamma_2 \end{bmatrix}$$

where γ_1 are the shadow prices of the generation constraints and γ_2 the constraints on the fixed load.

DC grid example, uncongested case (appendix):

$$\mathbf{L}(\mathbf{x}, \mathbf{P}, \lambda) = f_1(P_1) + f_2(P_2) + f_3(P_3) + \lambda_1 g_1(\mathbf{x}, \mathbf{P}) + \lambda_2 g_2(\mathbf{x}, \mathbf{P}) + \lambda_3 g_3(\mathbf{x}, \mathbf{P})$$

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} U_1 \\ U_2 \\ U_3 \end{pmatrix} \quad \mathbf{P} = \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} \quad \lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix}$$

Load flow equation for node 1:

$$P_1 = P_{12} + P_{13} = I_{12} U_1 + I_{13} U_1 = \left(\frac{U_1 - U_2}{R_{12}} \right) U_1 + \left(\frac{U_1 - U_3}{R_{13}} \right) U_1$$

$$g_1(\mathbf{x}, \mathbf{P}) = -P_1 + \left[\frac{(U_1^2 - U_1 U_2)}{R_{12}} + \frac{(U_1^2 - U_1 U_3)}{R_{13}} \right] = 0$$

$$g_2(\mathbf{x}, \mathbf{P}) = -P_2 + \left[\frac{(U_2^2 - U_2 U_1)}{R_{12}} + \frac{(U_2^2 - U_2 U_3)}{R_{23}} \right] = 0$$

$$g_3(\mathbf{x}, \mathbf{P}) = -P_3 + \left[\frac{(U_3^2 - U_3 U_1)}{R_{13}} + \frac{(U_3^2 - U_3 U_2)}{R_{23}} \right] = 0$$

Derivate wrt λ , $\mathbf{x}$ and $\mathbf{P}$:

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow \mathbf{g}(\mathbf{x}, \mathbf{P}) = 0$$

$$\frac{\partial L}{\partial \mathbf{x}} = 0 \Rightarrow \frac{\partial L}{\partial \mathbf{x}} = \begin{pmatrix} \lambda_1 \frac{\partial g_1}{\partial U_1} + \lambda_2 \frac{\partial g_2}{\partial U_1} + \lambda_3 \frac{\partial g_3}{\partial U_1} \\ \lambda_1 \frac{\partial g_1}{\partial U_2} + \lambda_2 \frac{\partial g_2}{\partial U_2} + \lambda_3 \frac{\partial g_3}{\partial U_2} \\ \lambda_1 \frac{\partial g_1}{\partial U_3} + \lambda_2 \frac{\partial g_2}{\partial U_3} + \lambda_3 \frac{\partial g_3}{\partial U_3} \end{pmatrix} = 0$$

Define Jacobian matrix:

$$\frac{\partial \mathbf{g}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}, \frac{\partial g_1}{\partial x_2}, \frac{\partial g_1}{\partial x_3} \\ \frac{\partial g_2}{\partial x_1}, \frac{\partial g_2}{\partial x_2}, \frac{\partial g_2}{\partial x_3} \\ \frac{\partial g_3}{\partial x_1}, \frac{\partial g_3}{\partial x_2}, \frac{\partial g_3}{\partial x_3} \end{pmatrix}$$

$$\frac{\partial L}{\partial \mathbf{x}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^t \boldsymbol{\lambda} = 0$$

$$\frac{\partial L}{\partial \mathbf{P}} = 0 \Rightarrow$$

$$\begin{pmatrix} \frac{\partial f}{\partial P_1} \\ \frac{\partial f}{\partial P_2} \\ \frac{\partial f}{\partial P_3} \end{pmatrix} + \begin{pmatrix} \frac{\partial g_1}{\partial P_1}, \frac{\partial g_2}{\partial P_1}, \frac{\partial g_3}{\partial P_1} \\ \frac{\partial g_1}{\partial P_2}, \frac{\partial g_2}{\partial P_2}, \frac{\partial g_3}{\partial P_2} \\ \frac{\partial g_1}{\partial P_3}, \frac{\partial g_2}{\partial P_3}, \frac{\partial g_3}{\partial P_3} \end{pmatrix} \boldsymbol{\lambda} = 0$$

$$\text{and} \quad \begin{pmatrix} c_1 \\ c_2 \\ d_3 \end{pmatrix} + \begin{pmatrix} -\lambda_1 \\ -\lambda_2 \\ -\lambda_3 \end{pmatrix} = 0$$

because $\frac{\partial f}{\partial P}$ are marginal generation costs and $\frac{\partial g}{\partial P} = -1$

NB error in equation (10) in Ch7-Appendix in the book!

Like before:

$$\mathbf{g}(\mathbf{x}, \mathbf{P}) = 0$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^t \boldsymbol{\lambda} = 0$$

$$\boldsymbol{\lambda} = \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

Losses

$$I_{12} = (U_1 - U_2) \frac{1}{R} \quad P_{12} = U_1 I_{12}$$

$$Loss_{12} = R I_{12}^2 = R \left(\frac{P_{12}}{U_1} \right)^2$$

$$\frac{\partial Loss_{12}}{\partial P_{12}} = 2 \frac{R \cdot P_{12}}{U_1^2} = 2 \left(1 - \frac{U_2}{U_1} \right)$$

If we transmit a small quantity ΔP_1 from node 1, and the corresponding quantity arriving at node 2 is ΔP_2 , then ρ_{12} is given by:

$$\rho_{12} = \frac{\Delta P_1 - \Delta P_2}{\Delta P_1} = 2 \left(1 - \frac{U_2}{U_1} \right) \quad \frac{U_2}{U_1} = 1 - \frac{\rho_{12}}{2}$$

In the two-node system:

$$g_1(\mathbf{x}, \mathbf{P}) = -P_1 + \left[(U_1^2 - U_1 U_2) / R \right] = 0$$

$$g_2(\mathbf{x}, \mathbf{P}) = -P_2 + \left[(U_2^2 - U_2 U_1) / R \right] = 0$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^t \boldsymbol{\lambda} = 0 \Rightarrow \begin{aligned} \lambda_1 \left[(2U_1 - U_2) / R \right] + \lambda_2 \left[-U_2 / R \right] &= 0 \\ \lambda_1 \left[-U_1 / R \right] + \lambda_2 \left[(2U_2 - U_1) / R \right] &= 0 \end{aligned}$$

Take second equation:

$$\frac{\lambda_1}{\lambda_2} = \frac{(2U_2 - U_1) / R}{U_1 / R} = 2 \frac{U_2}{U_1} - 1 = 2 \left(1 - \frac{\rho_{12}}{2} \right) - 1 = 1 - \rho_{12}$$

$$\text{or } \rho_{12} = 1 - \frac{\lambda_1}{\lambda_2} = \frac{\lambda_2 - \lambda_1}{\lambda_2} \quad \text{and in general } \frac{\lambda_i}{\lambda_j} = 1 - \rho_{ij}$$

This is valid in a DC grid – in an AC grid losses can be found by an iterative procedure.

Three node example, uncongested case, calculations:

Bus admittance matrix:

$$\mathbf{Y} = \begin{bmatrix} 0,3 & -0,1 & -0,2 \\ -0,1 & 0,15 & -0,05 \\ -0,2 & -0,05 & 0,25 \end{bmatrix}$$

Current vector:

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \mathbf{Y}\mathbf{x} = \begin{bmatrix} 0,3U_1 & -0,1U_2 & -0,2U_3 \\ -0,1U_1 & 0,15U_2 & -0,05U_3 \\ -0,2U_1 & -0,05U_2 & 0,25U_3 \end{bmatrix}$$

This just says that the current injected in bus 1 equals the currents 1-2 plus 1-3.

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} I_1U_1 \\ I_2U_2 \\ I_3U_3 \end{bmatrix} = \begin{bmatrix} 0,3U_1^2 & -0,1U_2U_1 & -0,2U_3U_1 \\ -0,1U_1U_2 & 0,15U_2^2 & -0,05U_3U_2 \\ -0,2U_1U_3 & -0,05U_2U_3 & 0,25U_3^2 \end{bmatrix}$$

$$\begin{aligned} \mathbf{g}(\mathbf{x}, \mathbf{P}) = 0 \quad \Rightarrow \quad & 0,3U_1^2 - 0,1U_1U_2 - 0,2U_1U_3 - P_1 = 0 \\ & -0,1U_1U_2 + 0,15U_2^2 - 0,05U_2U_3 - P_2 = 0 \\ & -0,2U_1U_3 - 0,05U_2U_3 + 0,25U_3^2 - P_3 = 0 \end{aligned}$$

Choose voltage in reference bus 3: 60 kV.

Given load $P_3 = 30$ MW, $G_1 = 20$ MW (lowest cost, no grid constraints, max):

$$\begin{aligned} 0,3U_1^2 - 0,1U_1U_2 - 0,2 \cdot 60 \cdot U_1 - 20 &= 0 \\ -0,1U_1U_2 + 0,15U_2^2 - 0,05 \cdot 60 \cdot U_2 - P_2 &= 0 \\ -0,2 \cdot 60 \cdot U_1 - 0,05 \cdot 60 \cdot U_2 + 0,25 \cdot 60^2 + 30 &= 0 \end{aligned}$$

from last equation: $U_2 = 310 - 4U_1$

Substitute in first equation and reorder: $0,7U_1^2 - 43U_1 - 20 = 0$

Result:

$$U_1 = 61,89 \text{ kV}$$

$$U_2 = 62,44 \text{ kV}$$

$$\mathbf{x} = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} 61,89 \\ 62,44 \\ 60,00 \end{bmatrix} \text{ [kV]}$$

Using equation for P_2 : $P_2 = 11,05 \text{ MW}$

$$\text{and } P_{loss} = \sum_i P_{Gi} - Load = 31,05 - 30 = 1,05 \text{ MW}$$

Nodal prices:

Determine from $\left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right)^t \boldsymbol{\lambda} = 0$:

$$\begin{aligned}\frac{\partial \mathbf{g}}{\partial \mathbf{x}} &= \begin{bmatrix} 0.6U_1 - 0.1U_2 - 0.2U_3 & -0.1U_1 \\ -0.1U_2 & 0.3U_2 - 0.1U_1 - 0.05U_3 \\ -0.2U_3 & -0.05U_3 \end{bmatrix} \\ &= \begin{bmatrix} 18.89 & -6.189 \\ -6.244 & 9.543 \\ -12 & -3 \end{bmatrix}\end{aligned}$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right)^t \boldsymbol{\lambda} = \begin{bmatrix} 18.89\lambda_1 - 6.244\lambda_2 - 12.00\lambda_3 \\ -6.189\lambda_1 + 9.543\lambda_2 - 3.00\lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Results in $\lambda_1 = 14.247 \text{ øre/kWh}$
 $\lambda_3 = 15.148 \text{ øre/kWh}$

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} c_1 + \gamma_1 \\ c_2 + \gamma_2 \\ \gamma_3 \end{bmatrix} = \begin{bmatrix} 14,247 \\ 14,0 \\ 15,148 \end{bmatrix} \text{ [øre/kWh]}$$

Loss coefficients:

$$\rho_{12} = 1 - \frac{\lambda_1}{\lambda_2} = 1 - \frac{14,247}{14} = -1,8\% \quad \rho_{21} = +1,7\%$$

$$\rho_{13} = 1 - \frac{\lambda_1}{\lambda_3} = 1 - \frac{14,247}{15,148} = +5,9\% \quad \rho_{31} = -6,3\%$$

$$\rho_{23} = 1 - \frac{\lambda_2}{\lambda_3} = 1 - \frac{14}{15,148} = 7,6\% \quad \rho_{32} = -8,2\%$$

Flows:

$$P_{12} = U_1 I_{12} = U_1 \frac{U_1 - U_2}{R_{12}} = 61,89 \cdot \frac{61,89 - 62,44}{10} = -3,40 \text{ MW}$$

etc.

Add grid congestion

$$L(\mathbf{x}, \mathbf{P}) = f(\mathbf{P}) + \boldsymbol{\lambda}^t \mathbf{g}(\mathbf{x}, \mathbf{P}) + \boldsymbol{\eta}^t \mathbf{h}(\mathbf{x}) + \boldsymbol{\gamma}^t (\mathbf{P} - \mathbf{P}_{\max})$$

$$L(\mathbf{x}, \mathbf{P}, \boldsymbol{\lambda}) = f_1(P_1) + f_2(P_2) + f_3(P_3)$$

$$+ \lambda_1 g_1(\mathbf{x}, \mathbf{P}) + \lambda_2 g_2(\mathbf{x}, \mathbf{P}) + \lambda_3 g_3(\mathbf{x}, \mathbf{P})$$

$$+ \eta_1 h_1(\mathbf{x}) + \eta_2 h_2(\mathbf{x}) + \eta_3 h_3(\mathbf{x})$$

$$+ \gamma_1 (P_1 - P_{1,\max}) + \gamma_2 (P_2 - P_{2,\max}) + \gamma_3 (P_3 - P_{3,\max})$$

$$\frac{\partial L}{\partial \mathbf{x}} = \begin{pmatrix} \lambda_1 \frac{\partial g_1}{\partial U_1} + \lambda_2 \frac{\partial g_2}{\partial U_1} + \lambda_3 \frac{\partial g_3}{\partial U_1} \\ \lambda_1 \frac{\partial g_1}{\partial U_2} + \lambda_2 \frac{\partial g_2}{\partial U_2} + \lambda_3 \frac{\partial g_3}{\partial U_2} \\ \lambda_1 \frac{\partial g_1}{\partial U_3} + \lambda_2 \frac{\partial g_2}{\partial U_3} + \lambda_3 \frac{\partial g_3}{\partial U_3} \end{pmatrix} + \begin{pmatrix} \eta_1 \frac{\partial h_1}{\partial U_1} + \eta_2 \frac{\partial h_2}{\partial U_1} + \eta_3 \frac{\partial h_3}{\partial U_1} \\ \eta_1 \frac{\partial h_1}{\partial U_2} + \eta_2 \frac{\partial h_2}{\partial U_2} + \eta_3 \frac{\partial h_3}{\partial U_2} \\ \eta_1 \frac{\partial h_1}{\partial U_3} + \eta_2 \frac{\partial h_2}{\partial U_3} + \eta_3 \frac{\partial h_3}{\partial U_3} \end{pmatrix} = 0$$

$$\frac{\partial L}{\partial \mathbf{P}} = \begin{pmatrix} \frac{\partial f_1}{\partial P_1} \\ \frac{\partial f_2}{\partial P_2} \\ \frac{\partial f_3}{\partial P_3} \end{pmatrix} + \begin{pmatrix} \frac{\partial g_1}{\partial P_1}, \frac{\partial g_2}{\partial P_1}, \frac{\partial g_3}{\partial P_1} \\ \frac{\partial g_1}{\partial P_2}, \frac{\partial g_2}{\partial P_2}, \frac{\partial g_3}{\partial P_2} \\ \frac{\partial g_1}{\partial P_3}, \frac{\partial g_2}{\partial P_3}, \frac{\partial g_3}{\partial P_3} \end{pmatrix} \boldsymbol{\lambda} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{\gamma} = 0$$

NB error in equation (22) in Ch7-Appendix in the book!

Vector with 1's because derivative of $(P - P_{\max})$ wrt P is 1.

Results in:

$$\begin{pmatrix} c_1 \\ c_2 \\ 0 \end{pmatrix} + \begin{pmatrix} -\lambda_1 \\ -\lambda_2 \\ -\lambda_3 \end{pmatrix} + \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{pmatrix} = 0 \Rightarrow \boldsymbol{\lambda} = \begin{bmatrix} \mathbf{c} + \boldsymbol{\gamma}_1 \\ \boldsymbol{\gamma}_2 \end{bmatrix}$$

Little different

from book, more correct

Calculation for example:

Set limit $P_{31} = 21$ MW (book uses 350 A).

$$\mathbf{h}(\mathbf{x}) \leq 0 \Leftrightarrow \begin{bmatrix} |P_{13}| - P_{13,max} \\ |P_{12}| - P_{12,max} \\ |P_{23}| - P_{23,max} \end{bmatrix} \leq 0$$

In reality these are 2 x 3 constraints, one in each direction.

Have that $-P_{13,max} \leq P_{13} \leq +P_{13,max}$

Can equation as:

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} |0.2(U_3 - U_1)U_3| - P_{3,1max} \\ |0.1(U_1 - U_2)U_1| - P_{1,2max} \\ |0.05(U_2 - U_3)U_2| - P_{2,3max} \end{bmatrix}$$

Use $P_{3,1}$ in first equation because book refers power to node 3 and want to show that this gives same result.

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial h_{11}}{\partial U_1} & \frac{\partial h_{11}}{\partial U_2} \\ \frac{\partial h_{21}}{\partial U_1} & \frac{\partial h_{21}}{\partial U_2} \\ \frac{\partial h_{31}}{\partial U_1} & \frac{\partial h_{31}}{\partial U_2} \end{bmatrix} = \begin{bmatrix} 0.2U_3 & 0 \\ 0.2U_1 - 0.1U_2 & -0.1U_2 \\ 0 & 0.1U_1 - 0.05U_3 \end{bmatrix}$$

because U_3 is given and constant

Constraint only on P_{13} :

$$\frac{\partial \mathbf{h}^t}{\partial \mathbf{x}} \boldsymbol{\eta} = \begin{bmatrix} 0.2U_3\eta_{13} \\ 0 \end{bmatrix} \quad \text{book uses } \eta \text{ because there is only one.}$$

General equations:

$$\mathbf{g}(\mathbf{x}, \mathbf{P}) = 0 \quad (a)$$

$$\mathbf{h}(\mathbf{x}) = 0 \quad (b)$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^t \boldsymbol{\lambda} + \left(\frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right)^t \boldsymbol{\eta} = 0 \quad (c)$$

$$\boldsymbol{\lambda} = \begin{bmatrix} \mathbf{c} + \boldsymbol{\gamma}_1 \\ \boldsymbol{\gamma}_2 \end{bmatrix} \quad (d)$$

Here:

$$\begin{aligned} 0.3U_1^2 - 0.1U_2U_1 - 0.2 \cdot 60 \cdot U_1 - P_1 &= 0 \\ -0.1U_1U_2 + 0.15U_2^2 - 0.05U_2 \cdot 60 - P_2 &= 0 \quad \text{error in book!} \quad (a) \\ -0.2U_1 \cdot 60 - 0.05U_2 \cdot 60 + 0.25 \cdot 60^2 - (-30) &= 0 \end{aligned}$$

$$0.2(U_3 - U_1)U_3 = 21 \quad (b)$$

$$\begin{aligned} \lambda_1(0.6U_1 - 0.1U_2 - 0.2 \cdot 60) - 0.1U_2\lambda_2 - 0.2 \cdot 60\lambda_3 + 0.2U_3\eta_{13} &= 0 \\ -0.1U_1\lambda_1 + \lambda_2(-0.1U_1 + 0.3U_2 - 0.05 \cdot 60) - 0.05 \cdot 60\lambda_3 &= 0 \end{aligned} \quad (c)$$

$$\begin{aligned} \lambda_1 &= 12 \\ \lambda_2 &= 14 \end{aligned} \quad (d)$$

γ_1 and γ_3 are zero because $P_{1,\max}$ and $P_{2,\max}$ are not binding.

From (b) solve $U_1 = 61.75$ kV (U_3 given)

Then (a) become 3 linear equations with unknowns U_2 , P_1 , P_2 . Solve U_2 from last, then P_1 from first and P_2 from second.

Then last equation in (c) solves λ_3 , and finally first equation in (c) solves η .